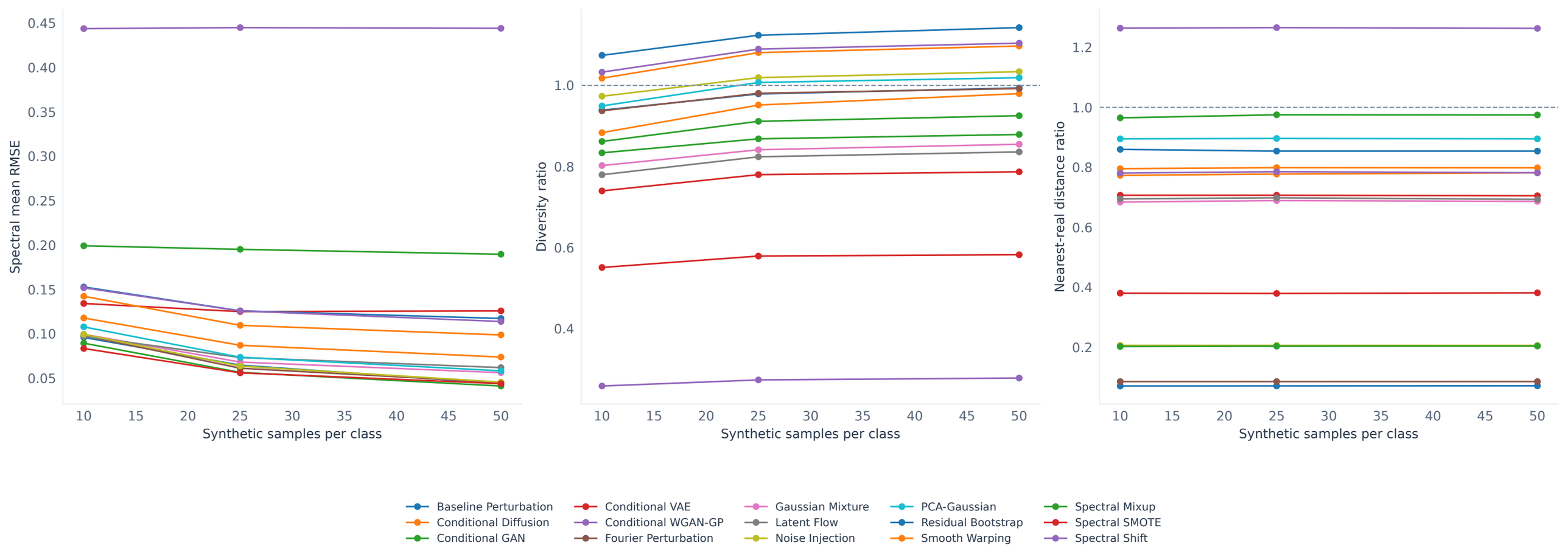
Spectral mean RMSE
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
10 15 20 25 30 35 40 45 50
Synthetic samples per class
Diversity ratio
1.0
0.8
0.6
0.4
10 15 20 25 30 35 40 45 50
Synthetic samples per class
Nearest-real distance ratio
1.2
1.0
0.8
0.6
0.4
0.2
10 15 20 25 30 35 40 45 50
Synthetic samples per class
Baseline Perturbation
Conditional Diffusion
Conditional GAN
Conditional VAE
Conditional WGAN-GP
Fourier Perturbation
Gaussian Mixture
Latent Flow
Noise Injection
PCA-Gaussian
Residual Bootstrap
Smooth Warping
Spectral Mixup
Spectral SMOTE
Spectral Shift